

TO OBSERVE THE EFFECTS OF METFORMIN HCl ON THE CARBOHYDRATE METABOLISM IN POLYCYSTIC OVARY SYNDROME

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ABSTRACT:

To observe the effects of metformin HCl on the carbohydrate metabolism in polycystic ovary syndrome a clinical trial was carried out on fifty infertile females having polycystic ovary syndrome with ages between 20-40 years. They were enrolled from the infertility clinic of a private hospital in Karachi and were given tablet metformin HCl 500 mg thrice daily for a period of three months. Fasting serum glucose (FSG) and fasting serum insulin (FSI) were done twice during the study period (at day zero and day ninety) to assess the effects on the carbohydrate metabolism. Significant reduction was found in both the FSG and FSI levels at day ninety with $P=0.001$. It is concluded that metformin has produced beneficial effects on the carbohydrate metabolism in patients having polycystic ovary syndrome. This is because of increase peripheral uptake and utilization of glucose being done by metformin which in turn increases the insulin sensitivity and reduces the insulin resistance in these patients.

Keywords: Polycystic Ovary Syndrome, Carbohydrate Metabolism, Fasting Serum Glucose, Fasting Serum Insulin, Metformin HCl.

INTRODUCTION

Polycystic ovary syndrome (PCOS) is the most common endocrinopathy of women of childbearing age affecting approximately 5% to 10% of women (Knochenhauer, 1998; Goldenberg *et al.*, 2005). PCOS is characterized by oligomenorrhoea, clinical and or biochemical hyperandrogenism and by polycystic ovaries (Kawadzki, 1992; Glueck, 2002; Rotterdam, 2003). Most but not all women with PCOS have fasting hyperinsulinemia with insulin resistance and high levels of insulin secretion (Meirow, 1995; Acien, 1999).

The consequences of polycystic ovary syndrome extend beyond the reproductive axis and women with the disorder are at substantial risk for the development of metabolic and cardiovascular abnormalities similar to those

that make up the metabolic syndrome (Glueck, 2003). Both the polycystic ovary syndrome and the metabolic syndrome share insulin resistance as a central pathogenetic feature. Polycystic ovary syndrome might thus be viewed as a sex specific form of the metabolic syndrome (NCEP, 2002) and the term "syndrome XX" has been suggested as an apt term to underscore this association (Sam, 2003).

Although the molecular mechanism(s) of insulin resistance in PCOS is unclear, excessive insulin independent serine phosphorylation of the β -subunit of the insulin receptor has put forward as a new mechanism for insulin resistance. This defect in insulin signaling pathway appears to be present in both the adipocyte and skeletal muscle, the primary target tissues of insulin action (Ehrmann, 2005). The association between a

Table-1
Baseline (Day-0) parameters of patients (n=50)

S. No.	Parameters	Range	Mean $\pm$ S.D.
1.	Fasting serum glucose (FSG) (mg/dl)	65-110	92.74 $\pm$ 13.0
2.	Fasting serum insulin (FSI) (μ U/ml)	10.4-51	20.6 $\pm$ 11.0

Table-2
Sub-groups of parameters and their percentage at Day-0 and Day-90 (n=50)

S.No.	Parameters	Subgroups		Day-0	Day-90
1.	Fasting serum glucose (FSG) (mg/dl)	(a)	< 100	31 (62%)	45 (90%)
		(b)	100-110	19 (38%)	05 (10%)
2.	Fasting serum insulin (FSI) (μ U/ml)	(a)	< 10	0	27 (54%)
		(b)	10-15	18 (36%)	16 (32%)
		(c)	> 15	32 (64%)	07 (14%)

disorder of carbohydrate metabolism and PCOS is not new and it was first described in 1921 (Achard and Thiers, 1921) and was called "the diabetes of bearded women" (diabete des femmes a barbe).

The pathway regulating carbohydrate metabolism is impaired in PCOS while insulin's action on steroidogenesis at the ovary is preserved despite the co-existence of peripheral insulin resistance (Willis and Franks, 1995). Insulin's action on steroidogenesis are mediated by pathways distinct from those which regulate the peripheral glucose disposal (Nestler *et al.*, 1998). It now appears that Ras-Raf-MEK (Mitogen Activated Protein Kinase) pathway is involved in the regulation of cell growth and metabolism whereas PI3-K (Phosphatidylinositol-3-Kinase) pathway is involved in glucose uptake and disposal (Cheatham *et al.*, 1994).

Metformin HCl is a biguanide anti-hyperglycemic agent which improves glucose tolerance in NIDDM subjects, lowering both basal and postprandial plasma glucose. It decreases hepatic glucose production, decreases intestinal absorption of glucose and improves insulin sensitivity (increases

peripheral glucose uptake and utilization). It does not produce hypoglycemia in either diabetic or non-diabetic subjects and does not cause hyperinsulinemia. With metformin therapy insulin secretion remains unchanged while fasting insulin levels and day long plasma insulin response may actually decrease. Metformin decreases the body weight and has a tendency to improve the lipid profile, particularly when baseline values are abnormally elevated (Diabeto Valens.com, 2002). Moreover by acting on the ovary and restoring normal ovarian activity metformin positively modulates the reproductive axis namely GnRH-LH episodic release and hence ovulation in females having PCOS (Genazzani *et al.*, 2004).

MATERIALS AND METHODS

A total of 50 infertile females having PCOS (polycystic ovary syndrome) were selected from the infertility clinic of a private hospital, Karachi. Eligibility criteria included women of reproductive age group with ages between 20-40 years having infertility, oligomenorrhoea, obesity, hirsutism, fasting hyperinsulinemia ($>10\mu$ U/ml) and fasting serum sugar level < 6.1 mmol/L (≤ 110 mg/dl) as ≥ 6.1 mmol/L is WHO diabetic criteria,

Table-3
Fasting serum glucose (FSG) at Day-0 and Day-90
(n=50)

Parameter	Day-0 Mean $\pm$ S.D.	Day-90 Mean $\pm$ S.D.	P-value
Fasting serum glucose (mg/dl)	92.74 $\pm$ 13.0	86 $\pm$ 8.7	0.001

Table-4
Change in fasting serum glucose (FSG) from Day-0 to Day-90
(n=50)

Decrease in fasting serum glucose		Increase in fasting serum glucose		No change	
No.	%	No.	%	No.	%
31	62	19	38	-	-

2000. Consent were obtained from all study participants before they were enrolled in the study. Preliminary data, date of follow up visit and laboratory investigations of each patient were recorded on a specially designed proforma.

Study Design

This clinical trial was done for a period of three months on each patient. Following a thorough history and physical examination the patients were given tablet metformin HCl 500 mg thrice daily for a period of three months. Metformin HCl was started initially once a day for one week and then increased to thrice daily in the next week. The starting week is not included in the study period. Fasting serum glucose and insulin were done twice during the study i.e. at day zero and at day ninety.

Analytical Methods

Fasting serum glucose was determined by enzymatic colorimetric method in which serum glucose is oxidized by glucose oxidase enzyme to gluconic acid and H_2O_2 . The formed H_2O_2 then reacts under catalytic action of peroxidase with hydroxybenzoic acid and 4-aminoantipyrine to yield a red violet

quinoneimine dye. The quantity of this dye is proportional to the amount of glucose present in the sample.

Fasting serum insulin levels were measured by using IMX insulin reagent. Insulin assay is based on microparticle enzyme immunoassay (MEIA) technology. The IMX system has been designed to perform immunoassays using fluorogenic enzyme substrates and fluorescence polarization techniques. This enzyme immunoassays procedure typically uses a coated submicron microparticle as the means by which the analyte to be measured is captured for analysis. Assays which use this method are called the microparticle enzyme immunoassays (MEIA).

RESULTS

Fasting serum glucose in mg/dl was between 65-110 mg/dl equivalent to a mean $\pm$ SD of 92.74 $\pm$ 13. Fasting serum insulin was between 10.4 μ U/ml to 51 μ U/ml that is a mean $\pm$ SD of 20.6 $\pm$ 11 (Table 1).

For observing the fasting serum glucose

Table-5
fasting serum insulin (FSI) at Day-0 and Day-90
(n=50)

Parameter	Day-0 Mean $\pm$ S.D.	Day-90 Mean $\pm$ S.D.	P-value
Fasting serum insulin (μ U/ml)	20.6 $\pm$ 11.0	9.80 $\pm$ 5.6	0.001

Table-6
Change in fasting serum insulin (FSI) from Day-0 to Day-90
(n=50)

Decrease in fasting serum insulin		Increase in fasting serum insulin		No change	
No.	%	No.	%	No.	%
50	100	-	-	-	-

(FSG) patients were divided into two groups according to inclusion criteria of FSG $\leq$ 110 mg/dl. Group (a) with FSG level of $<$ 100 mg/dl had initially 31(62%) patients which increased to 45 (90%) after treatment while group (b) with FSG level between 100-110 mg/dl had 19 (38%) patients whose number decreased to only 5 (10%) after treatment. For fasting serum insulin (FSI) in μ U/ml patients were divided into three groups (a) FSI $<$ 10, (b) FSI between 10-15 and (c) FSI $>$ 15 μ U/ml according to the inclusion criteria. Before treatment group (a) did not had any patients but following a 90 days treatment 27 (54%) patients fall into this group. Group (b) had 18 (36%) patients initially which decreased to 16 (32%), patients after treatment. Group (c) had 32 (64%) patients which decreased to only 7 (14%) after treatment (Table-2).

A decrease was observed in fasting serum glucose from a mean $\pm$ SD of 92.74 $\pm$ 13 to 86 $\pm$ 8.7 mg/dl from day-0 to day-90. It was significant with P-value of 0.001 (Table-3). 31(62%) patients showed decrease in FSG while 19 (38%) showed an increase in fasting serum glucose (Table-4).

Fasting serum insulin showed a significant decrease with P-value of 0.001 from 20.6 $\pm$ 11.0 μ U/ml to 9.80 $\pm$ 5.6 μ U/ml (Table-5). All the 50 patients showed a decrease in FSI making a percentage of 100 (Table-6).

DISCUSSION

The polycystic ovary syndrome is one of the most common hormonal disorders affecting women. As a syndrome it has multiple components-reproductive, metabolic and cardiovascular – with health implications across the life span. Insulin resistance of the polycystic ovary syndrome appears to impart an increased risk of glucose intolerance, diabetes and lipid abnormalities. A better understanding of the pathogenesis of insulin resistance that is associated with complications of the polycystic ovary syndrome has led to novel therapies – chiefly insulin lowering medications (Ehrmann, 2005).

Fasting serum glucose level reduced significantly in our study from 92.74 $\pm$ 13.0 mg/dl to 86 $\pm$ 8.37 mg/dl which is coinciding with the study of Morin-Papunen *et al.* (2000). He observed significant reduction in fasting

serum glucose level of 8 PCOS patients from 5.2 ± 0.1 (92.82 mg/dl) to 4.9 ± 0.1 mmol/L (87.5 mg/dl) [$\pm$ mmol/L by 0.056 conversion factor to get mg/dl] after 3 months of metformin therapy. Whereas Marca *et al.* (2002) in 15 PCOS women after 35-40 days of metformin therapy and Kolodziejczyk (2000) in 35 women after 12 weeks of metformin therapy found non-significant reduction in the fasting glucose level. On the contrary Nestler (1998) observed non-significant increase in the fasting glucose level from 78 ± 3 mg/dl to 81 ± 3 mg/dl after 35 days of treatment. Short period of this study is probably related to these results.

As regards the fasting serum insulin level we have observed a significant reduction in the mean fasting serum insulin level from 20.6 ± 11.0 to 9.8 ± 5.6 μ U/ml which is matching with the studies of Nestler (1996), Kolodziejczyk (2000), Morin-Papunen (2000) Marca (2002) whereas Moghetti (2000) in a study of 23 caucasian women for 6 months with metformin 500 mg thrice daily found non-significant decrease in fasting serum insulin level from 15.2 ± 4.6 to 10.2 ± 2.2 – U/ml. Long duration of study of 6 months and Caucasian habitat of the females may be attributed to these results.

CONCLUSION

In our study insulin lowering drug metformin HCl has exerted beneficial effects on the carbohydrate metabolism in PCOS females by significantly reducing the fasting serum glucose and fasting serum insulin levels following a three months therapy with the above-mentioned biguanide.

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